

BOSTON UNIVERSITY
PROMYS 2023

Exploring Sums of Squares and r -gonal Numbers

Anastasia Lee, Summer Li, Ladina Moor, Liz Zhao

Mentored by Akiva Weinberger and Supervised by David Fried

August 28, 2023

Contents

1	Introduction	3
2	Distinct Distances Between Pairs of Lattice Points	4
2.1	Numerical Examples	4
2.2	Code	5
3	Number of Lattice Points in a Closed Disk	6
3.1	Numerical Examples	6
3.2	Code	8
4	Decomposing Numbers into Sums of Two Squares	9
4.1	Numerical Examples	13
4.2	Code	14
5	Decomposing Numbers into Sums of Three Squares	16
5.1	Code	18
6	Decomposing Numbers into Sums of Four Squares	19
6.1	Code	20
7	r-gonal Numbers	21
7.1	General Findings	21
7.2	Triangular Numbers	22
7.3	Pentagonal Numbers	25
7.4	Hexagonal Numbers	26
7.5	Numerical Examples	27
8	Conclusion	28

1 Introduction

There are simple but scintillating questions about decomposing natural numbers into sums of squares. Consider a 5×5 grid. We know from our experiences that we can find both horizontal and vertical distances of 5 between pairs of lattice points, but is it possible to find other such pairs such that the distance between them is 5? Asking ourselves this question is equivalent to finding a decomposition of 25 into a sum of two positive squares. We can write this as the expression

$$5^2 = 3^2 + 4^2$$

This reminds us of the Pythagorean triple (3,4,5). A question that naturally arises from this numerical example is which numbers can be decomposed into perfect squares? Then we posit the question, in how many ways can we decompose such a number? Which leads us to ask about when can we decompose numbers into sums of two squares versus sums of three squares or even sums of four squares? This paper describes some of these ideas, but is by no means comprehensive.

We begin by providing some crucial definitions which will be fundamental to understanding our topic of discussion.

Definition 1.1. $\mathbb{N} := \mathbb{Z}^+ \cup \{0\}$. Our motivation for defining the set of natural numbers in this way is to account for all cases in which we can decompose a number into a sum of squares that includes zero as a summand.

Definition 1.2. We define D to be the set of distinct distances in an $n \times n$ grid. Thus, $\forall k \in D, \exists a, b \in \mathbb{N}$, where $a, b \leq n$ such that $k = \sqrt{a^2 + b^2}$.

Definition 1.3. We define an $n \times n$ grid as the collection of points (a, b) such that $a, b \in \mathbb{Z}$, $0 \leq a, b \leq n$.

Definition 1.4. We define $d(k)$ to be the multiplicity of a distance $k \in D$. Since $k \in D$, $d(k) \geq 1$.

Definition 1.5. $\mathbb{Z}[i] := \{a + bi : a, b \in \mathbb{Z}\}$, where $i = \sqrt{-1}$.

Definition 1.6. $\forall \alpha = a + bi \in \mathbb{Z}[i], N(\alpha) = a^2 + b^2$.

Definition 1.7. $\forall r \in \mathbb{Z}^+, R_n$ is the n th r -gonal number.

Definition 1.8. Fundamental Theorem of Arithmetic for Gaussian Integers: $\forall a, b, c \in \mathbb{Z}[i]$, if $a|bc$ and $(a, b) = 1$ then $a|c$. A corollary of this theorem is $\forall p, a, b \in \mathbb{Z}[i]$, if p is prime and $p|ab$, then either $p|a$ or $p|b$.

2 Distinct Distances Between Pairs of Lattice Points

2.1 Numerical Examples

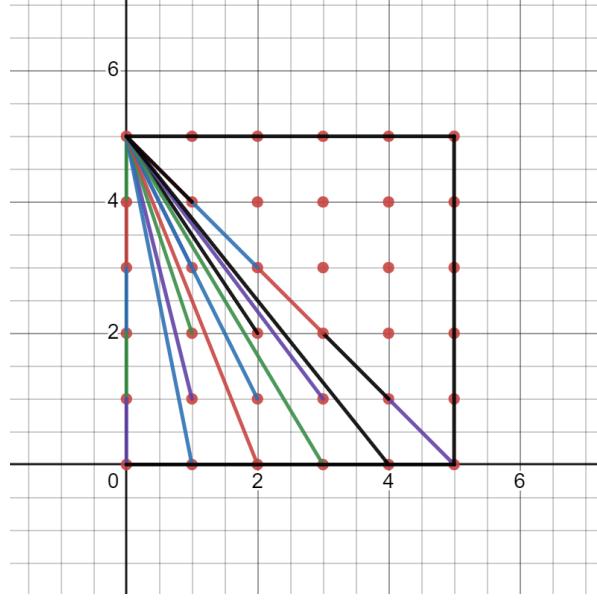


Figure 1: Distinct distances between pairs of lattice points on a 5×5 grid

Theorem 2.1

The number of distinct distances in an $n \times n$ grid is

$$\frac{(n+1)(n+2)}{2} - \sum_{k \in D} d(k) - 1. \quad (1)$$

Proof. Without loss of generality, we can plot the $n \times n$ grid on the Cartesian plane where the coordinates of the vertices are $(0,0)$, $(0,n)$, $(n,0)$, (n,n) . We can generate all possible distances by starting at a vertex of the $n \times n$ grid, say the origin. Observe that any distance in the $n \times n$ grid that does not include the origin as an endpoint can undergo a rigid transformation so that one endpoint is at $(0,0)$ and the other endpoint is on the grid. Since the distance between $(0,0)$ and an arbitrary point (a,b) is the same as the distance from $(0,0)$ to (b,a) , we can split the grid in half using the diagonal from $(0,n)$ to $(n,0)$. We can then count the distances from $(0,0)$ to any lattice point inside and on the boundary of the triangle using the following expression:

$$(n+1) + (n) + (n-1) + \cdots + 1 = \frac{(n+2)(n+1)}{2}$$

However, we must consider overcounting. There are some distances with a multiplicity

greater than 1. For example,

$$5 = \sqrt{5^2 + 0^2} = \sqrt{3^2 + 4^2} \quad \text{and} \quad \sqrt{50} = \sqrt{7^2 + 1^2} = \sqrt{5^2 + 5^2}.$$

Therefore, we subtract $\sum_{k \in D} d(k) - 1$. This yields (1) thus completing the proof. $\square$

Number of Distinct Distances in an $n \times n$ grid	
n	$ D $
0	1
1	3
2	6
3	10
4	15
5	20
6	27

2.2 Code

The code listed below was used to efficiently count the multiplicity of distinct distances in D for various $n \times n$ grids.

```

1 import java.util.*;
2 public class MyClass {
3     public static void main(String args[]) {
4         Scanner in = new Scanner(System.in);
5         //gets input from user - dimension of grid
6         int n = in.nextInt();
7         //creates a set to store distances
8         Set<Double> distances = new HashSet<Double>();
9         //loops over all horizontal and vertical lengths
10        for (int i = 0; i <= n; i++) {
11            for (int j = 0; j <= n; j++) {
12                //adds length of hypotenuse to the set
13                distances.add(Double.valueOf(i*i+j*j));
14            }
15        }
16        //output the size of the set
17        System.out.println(distances.size());
18    }
19 }

```

Listing 1: Counting the Number Distinct Distances Between Pairs of Lattice Points

3 Number of Lattice Points in a Closed Disk

Definition 3.1. A closed disk is a circle that contains all lattice points in and on the circle.

Closed Disks of Radius (r)

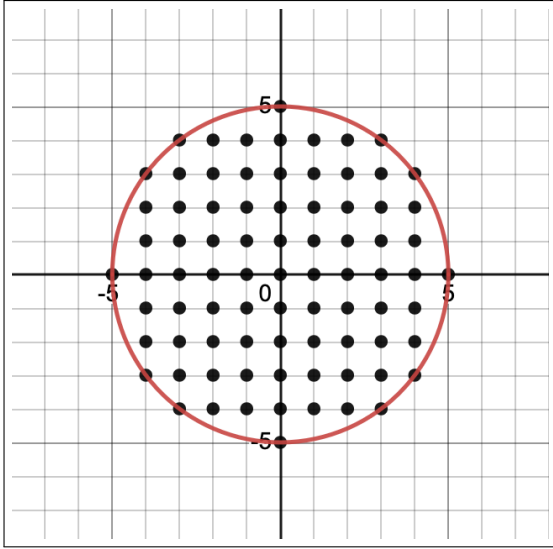


Figure 2: Closed Disk of Radius 5

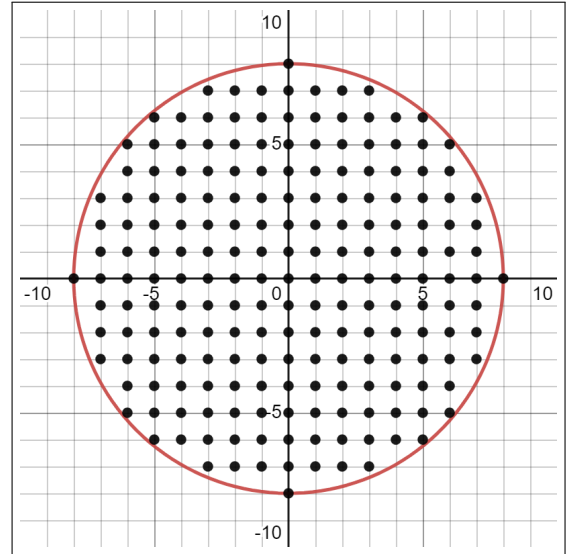


Figure 3: Closed Disk of Radius 8

3.1 Numerical Examples

Here are some examples of the number of lattice points in a closed disk of radius r . We can also relate these values to the area of the circle with radius r , using the equation for the area of a circle: $A = \pi r^2$.

Radius (r)	Num. of Lattice Points	Area of Circle
1	5	3.14159
π	29	31.006
6.85545	145	147.654855
7	149	153.93804
158	78413	78426.719
256	205861	205887.41
1000	3141549	3141592.65

Remark 3.2. The number of lattice points in a closed disk is an estimation of the area of the circle, and the accuracy increases as the radius increases.

Theorem 3.3

The number of lattice points in a closed disk with radius r is:

$$(2\lfloor r \rfloor + 1) + 2 \sum_{n=1}^{\lfloor r \rfloor} \left(2\lfloor \sqrt{r^2 - n^2} \rfloor + 1 \right) \quad (2)$$

Proof. Without loss of generality, let the closed disk be centered at $(0,0)$. Then, the number of lattice points on the diameter of the closed disk along the x -axis is $2\lfloor r \rfloor + 1$, where there are $\lfloor r \rfloor$ ¹ number of lattice points to the left and right of the y -axis and single lattice point at the on center of the closed disk. We can then find the remaining number of lattice points in the closed disk that are above the x -axis, by finding the sum of all lattice points in the closed disk and on each line of the form:

$$y = n \quad \text{where } 0 < n \leq \lfloor r \rfloor$$

To do this, we must find the corresponding x -coordinates on the circle for each line $y = n$. At these values, there exists a horizontal secant line that is symmetric about the y -axis and has a length of $2\sqrt{r^2 - n^2}$. We can then say that there are a total of $2\lfloor \sqrt{r^2 - n^2} \rfloor$ ¹ number of lattice points on the secant line excluding the point located on the y -axis which we account for in adding one to the expression. Thus, we find that there are $2\lfloor \sqrt{r^2 - n^2} \rfloor + 1$ lattice points on each line $y = n$. We can then write that the total number of lattice points on every secant line above the x -axis of the closed disk is:

$$\sum_{n=1}^{\lfloor r \rfloor} \left(2\lfloor \sqrt{r^2 - n^2} \rfloor + 1 \right)$$

Because our circle is centered at $(0,0)$ it is symmetric about the x -axis. Thus, we can say that there are an equal number of lattice points in the closed disk above and below the x -axis. This yields a total number of (2) lattice points . $\square$

¹We take the floor of r to account for all radii that are not integral values.

3.2 Code

The code listed below was used to efficiently generate the number of lattice points in a closed disk of radius r .

```
1 import java.util.*;
2 public class InsideCircle {
3     public static void main(String[] args) {
4         Scanner in = new Scanner(System.in);
5         //gets input from user - radius of circle
6         double r = in.nextDouble();
7         int ans = 0;
8         //loops over all integers between 1 and floor(r)
9         for (int i = 1; i <= Math.floor(r); i++) {
10             //number of lattice points on the line y = i
11             ans += 2*Math.floor(Math.sqrt(r*r-i*i)) + 1;
12         }
13         ans *= 2;
14         //add points on x-axis
15         ans += 2*Math.floor(r) + 1;
16         //output total number of lattice points
17         System.out.println(ans);
18     }
19 }
```

Listing 2: Counting the Number of Lattice Points in a Closed Disk

4 Decomposing Numbers into Sums of Two Squares

Example 4.1

Is it possible to write 3929 as a sum of two squares? How about 2857? 3527?

- $3929 = 35^2 + 52^2$
- $2857 = 16^2 + 51^2$
- 3527 cannot be written as the sum of two squares.

Proposition 4.2

$\forall n \in \mathbb{N}$, if $n \equiv 3 \pmod{4}$, then $\nexists a, b \in \mathbb{N}$ such that $n = a^2 + b^2$.

Proof. Let $n \in \mathbb{N}$ satisfy $n \equiv 3 \pmod{4}$. Suppose for the sake of contradiction, $\exists a, b \in \mathbb{N}$ such that $n = a^2 + b^2$. Taking the equation modulo 4, we obtain the congruence

$$3 \equiv a^2 + b^2 \pmod{4}.$$

However, the only perfect squares in $\mathbb{Z}_4$ are 0 and 1:

$$0^2 \equiv 0 \pmod{4}$$

$$1^2 \equiv 1 \pmod{4}$$

$$2^2 \equiv 0 \pmod{4}$$

$$3^2 \equiv 1 \pmod{4}.$$

Therefore, $\forall a, b \in \mathbb{N}$, $a^2 + b^2 \in \{0, 1, 2\} \pmod{4}$. This completes the proof. $\square$

Example 4.3

Can the numbers $11,225,153 = 3929 \cdot 2857$ and $13,857,583 = 3929 \cdot 3527$ be written as a sum of two squares?

- $11,225,153 = 3929 \cdot 2857 = 953^2 + 3212^2 = 2092^2 + 2617^2$
- $13,857,583 = 3929 \cdot 3527$ cannot be written as the sum of two squares.

For more examples reference our [table](#) below on counting the number of ways to decompose an integer into sums of two perfect squares

Theorem 4.4

$\forall x, y \in \mathbb{N}$, $\exists m, n \in \mathbb{N}$ such that $xy = m^2 + n^2$ if $\exists a, b, c, d \in \mathbb{N}$ $x = a^2 + b^2$ and $y = c^2 + d^2$.

Proof. Consider $x, y \in \mathbb{N}$, with $x = a^2 + b^2$ and $y = c^2 + d^2$ for some $a, b, c, d \in \mathbb{N}$. Then

$$\begin{aligned} xy &= (a^2 + b^2)(c^2 + d^2) \\ &= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 \\ &= (a^2c^2 - 2abcd + b^2d^2) + (a^2d^2 + 2abcd + b^2c^2) \\ &= (ac - bd)^2 + (ad + bc)^2. \end{aligned}$$

If $ac - bd \in \mathbb{N}$, then we can set $m = ac - bd$ and $n = ad + bc$. If $ac - bd \notin \mathbb{N}$, then $bd - ac \in \mathbb{N}$, so since $(ac - bd)^2 = (-(ac - bd))^2$, we can set $m = bd - ac$ and $n = ad + bc$. Either way, the proof is complete. $\square$

Remark 4.5. We can notice that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2 = (ac + bd)^2 + (ad - bc)^2$, showing that in the general case, there is more than one way to write a product of two sums of two squares as a sum of two squares.

Remark 4.6. The converse of [Theorem 4.4](#) is not true if for some $x, y \in \mathbb{N}$, $\exists m, n \in \mathbb{N}$ such that $xy = m^2 + n^2$, we do not necessarily have that $\exists a, b, c, d \in \mathbb{N}$ $x = a^2 + b^2$ and $y = c^2 + d^2$. A counterexample is that $3 \cdot 6 = 18 = 3^2 + 3^2$, but 3 and 6 cannot be written as a sum of two squares.

Let us consider some prime integers $a, b \in \mathbb{N}$, such that both a and b can not be decomposed into sums of two squares. Suppose for the sake of contradiction that $ab \in \mathbb{N}$, $ab = x^2 + y^2$.

Remark 4.7. $\forall n \in \mathbb{N}$, $\exists a, b \in \mathbb{N}$ such that $n = a^2 + b^2$ iff $\exists z \in \mathbb{Z}[i]$ such that $N(z) = n$. Moreover, $\forall x, y \in \mathbb{Z}[i]$, $N(xy) = N(x)N(y)$ by the proof of [Proposition 4.2](#)

This connection to the norms of Gaussian integers is crucial to understanding what numbers can be expressed as a sum of two squares. We further exploit this fact below.

Definition 4.8. $p \in \mathbb{Z}[i]$ is prime iff $\forall a, b \in \mathbb{Z}[i]$ that satisfy $p = ab$, we have that exactly one of $N(a)$, $N(b)$ is equal to 1 in $\mathbb{Z}$.

Lemma 4.9

If an odd integer prime p satisfies $p \equiv 1 \pmod{4}$, then $\exists x \in \mathbb{Z}_p$ such that $x^2 \equiv -1 \pmod{p}$.

Proof. By Wilson's Theorem, $(p-1)! \equiv -1 \pmod{p}$. If $p \equiv 1 \pmod{4}$, then

$$\begin{aligned}
 (p-1)! &\equiv 1 \cdot 2 \cdots \left(\frac{p-1}{2}\right) \cdot \left(\frac{p+1}{2}\right) \cdots (p-1) \pmod{p} \\
 &\equiv 1 \cdot 2 \cdots \left(\frac{p-1}{2}\right) \cdot \left(-\frac{p-1}{2}\right) \cdots (-1) \pmod{p} \\
 &\equiv \left(\left(\frac{p-1}{2}\right)!\right)^2 (-1)^{\frac{p-1}{2}} \pmod{p} \\
 &\equiv 1 \pmod{p}.
 \end{aligned}$$

Thus, there exists a solution to $x^2 \equiv -1 \pmod{4}$, completing the proof. $\square$

Lemma 4.10

Given a prime $p \in \mathbb{Z}^+$, p is not prime in $\mathbb{Z}[i]$ if $p \equiv 1 \pmod{4}$.

Proof. By Lemma 4.9, if $p \equiv 1 \pmod{4}$, then $\exists x \in \mathbb{Z}_p$ such that $x^2 \equiv -1 \pmod{p}$, or equivalently, $x^2 + 1 \equiv 0 \pmod{p}$. In $\mathbb{Z}[i]$, $x^2 + 1 = (x+i)(x-i)$, so we must have $p \mid (x+i)(x-i)$.

Now assume for the sake of contradiction that p is prime in $\mathbb{Z}[i]$. Then from the Fundamental Theorem of Arithmetic for Gaussian Integers, either $p \mid (x+i)$ or $p \mid (x-i)$. But this implies that $\exists a+bi \in \mathbb{Z}[i]$ such that $p(a+bi) = x \pm i$, so $pb = \pm 1$. Since $p > 2$, there is no such integer b . Thus, we have a contradiction, and p is not prime in $\mathbb{Z}[i]$. $\square$

Lemma 4.11

If $p \in \mathbb{Z}$ is not prime in $\mathbb{Z}[i]$, then $\exists a, b \in \mathbb{N}$ such that $p = a^2 + b^2$.

Proof. By Definition 4.8, if p is not prime in $\mathbb{Z}[i]$, then $\exists d, k \in \mathbb{Z}[i]$ such that $p = dk$ with $N(d), N(k) > 1$. Now, from Theorem 4.4, $N(p) = N(dk) = N(d)N(k)$, and since $p \in \mathbb{Z}$, $N(p) = p^2$. The only factors of p^2 in $\mathbb{Z}$ are $1, p, p^2$, but because neither of $N(d), N(k)$ are equal to 1, we must have $N(d) = N(k) = p$. By the definition of a norm (Definition 1.6), this means that $\exists a, b \in \mathbb{Z}$ such that $p = a^2 + b^2$, completing the proof. $\square$

Theorem 4.12

Given any odd integer prime p , $\exists a, b \in \mathbb{N}$ such that $p = a^2 + b^2$ iff $p \equiv 1 \pmod{4}$.

Proof. An odd prime integer p must satisfy either $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$. From Proposition 4.2, if $p \equiv 3 \pmod{4}$, $\nexists a, b \in \mathbb{N}$ such that $p = a^2 + b^2$. Thus, what remains to be proven is that if $p \equiv 1 \pmod{4}$, then $\exists a, b \in \mathbb{N}$ such that $p = a^2 + b^2$.

By [Lemma 4.10](#), if $p \equiv 1 \pmod{4}$, then p is not prime in $\mathbb{Z}[i]$, and by [Lemma 4.11](#), this implies that $\exists a, b \in \mathbb{N}$ such that $p = a^2 + b^2$. Thus the proof is complete. $\square$

Lemma 4.13

Given $p, a, b \in \mathbb{N}$, p a prime satisfying $p \equiv 3 \pmod{4}$ and $p|(a^2 + b^2)$, we have $p|a$ and $p|b$.

Proof. From the contrapositive of [Lemma 4.11](#), we have that if $\nexists c, d \in \mathbb{N}$ such that $p = c^2 + d^2$, then p is prime in $\mathbb{Z}[i]$. By [Proposition 4.2](#), this is true for all $p \equiv 3 \pmod{4}$. Then by the Fundamental Theorem of Arithmetic for Gaussian Integers, $p|(a^2 + b^2) = (a + bi)(a - bi)$ implies $p|(a + bi)$ or $p|(a - bi)$. Either way, we must have $p|a$ and $p|b$, completing the proof. $\square$

Theorem 4.14

$\forall n \in \mathbb{N}$, $\exists x, y \in \mathbb{N}$ such that $n = x^2 + y^2$ iff all prime factors p of n satisfying $p \equiv 3 \pmod{4}$ are raised to even powers.

Proof. Consider some $a, b \in \mathbb{N}$, such that $n = a^2b$, where b is the square-free part of n . If every prime factor p of n satisfying $p \equiv 3 \pmod{4}$ is raised to an even power, then $p|a^2$ and $p \nmid b$ for all p . Then it follows that for all primes q such that $q|b$, either $q = 2$ or $q \equiv 1 \pmod{4}$. We can check that 2 can be written as a sum of two squares, namely $2 = 1^2 + 1^2$. Furthermore, by [Theorem 4.12](#), if $q \equiv 1 \pmod{4}$, then q can be written as a sum of two squares. So, b is a product of sums of squares, meaning that for some $c, d \in \mathbb{N}$ $b = c^2 + d^2$. We then have

$$\begin{aligned} n &= a^2(c^2 + d^2) \\ &= (ac)^2 + (ad)^2. \end{aligned}$$

Thus, n is a sum of two squares.

Conversely, we can prove that if $\exists x, y \in \mathbb{N}$ such that $n = x^2 + y^2$, then there are no prime factors $p \equiv 3 \pmod{4}$ dividing b , and thus all primes $p \equiv 3 \pmod{4}$ dividing n are raised to an even power. We proceed by inducting on n . For the base case, we can see that the statement holds when $n = 0$, since there are no prime factors $p \equiv 3 \pmod{4}$.

Now for the inductive hypothesis, assume that for all $k < n$, the statement is true. If n has no prime factors $p \equiv 3 \pmod{4}$, then we are done, so suppose $\exists p \equiv 3 \pmod{4}$ such that $p|n$. If $n = x^2 + y^2$, then by [Lemma 4.13](#), $p|x$ and $p|y$. Thus, we can write $x = px_1$ and $y = py_1$ for some $x_1, y_1 \in \mathbb{Z}$. Therefore, we get

$$n = x^2 + y^2 = (px_1)^2 + (py_1)^2 = p^2(x_1^2 + y_1^2).$$

Since $x_1^2 + y_1^2 < x^2 + y^2$, by our inductive hypothesis $x_1^2 + y_1^2 = a_1^2b$ with no primes $p \equiv 3 \pmod{4}$ dividing b . So, $n = x^2 + y^2 = p^2a_1^2b = (pa_1)^2b$ also satisfies the condition, completing the proof. $\square$

4.1 Numerical Examples

Decomposing Prime Integer Powers p^n into Sums of Two Perfect Squares (Note: $p \equiv 1 \pmod{4}$)			
p^n			$d(p^n)$
5^1	13^1	101^1	1
5^2	13^2	101^2	2
5^3	13^3	101^3	2
5^4	13^4	101^4	3
5^5	13^5	101^5	3
5^6	13^6	101^6	4
5^7	13^7	101^7	4
5^8	13^8	101^8	5
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Conjecture 4.15. We assert, yet have not yet proven, that $\forall p \in \mathbb{Z}^+$ such that $p \equiv 1 \pmod{4}$, p^n can be written as the sum of two squares in $\left\lfloor \frac{n}{2} \right\rfloor + 1$ ways.

Counting the Number of Ways to Decompose an Integer n into Sums of Two Perfect Squares			
n	$d(n)$	n	$d(n)$
2	1	$13 \cdot 17$	2
5	1	$13 \cdot 17 \cdot 5^2$	6
13	1	$13 \cdot 5^2$	3
17	1	$13 \cdot 5^2 \cdot 29$	6
25	2	$5^3 \cdot 13^2$	6
29	1	5^3	2
41	1	$13 \cdot 5^2 \cdot 265$	8
5^2	2	$5 \cdot 13 \cdot 17 \cdot 29 \cdot 41$	16
$5^2 \cdot 2$	2	$5 \cdot 13 \cdot 17 \cdot 29 \cdot 41 \cdot 53$	32
$5 \cdot 17$	2	$5^2 \cdot 17^2$	5
$5^2 \cdot 17$	2	$5^3 \cdot 17^3$	8
$5 \cdot 13 \cdot 17$	4	$5^4 \cdot 13^4$	13
$5^3 \cdot 13 \cdot 17$	8	$5 \cdot 13$	2
5^4	3	5^6	4
$5 * 13 * 17$	4	$5 * 13 * 17 * 29$	8

Conjecture 4.16. Let n be a natural number and let x denote the number of primes in the prime factorization of n that are $1 \pmod{4}$. If the power of these primes is 1, then there are 2^{x-1} ways n can be decomposed as a sum of two squares.

4.2 Code

The code listed below was used to efficiently generate the number of decompositions into sums of two squares for an integer n

```

1 import java.util.*;
2 public class TwoSquares {
3     public static void main(String args[]) {
4         Scanner in = new Scanner(System.in);
5         //gets input from user
6         int x = in.nextInt();
7         //creates ArrayList to store decompositions
8         ArrayList<String> squares = new ArrayList<String>();
9         //loops over integers less than sqrt(x)

```

```

10     for (int i = 0; i < Math.sqrt(x); i++) {
11         //calculates the number for which  $x = i^2 + y^2$ 
12         double y = Math.sqrt(x-i*i);
13         //checks if y is an integer
14         if (Math.floor(y) == y && i <= y) {
15             //adds decomposition to the ArrayList
16             squares.add(i + " " + Math.round(y));
17         }
18     }
19     //outputs the number of decompositions and each decomposition
20     System.out.println(squares.size());
21     for (String s:squares) {
22         System.out.println(s);
23     }
24 }
25 }

```

Listing 3: Counting the Number of Ways to Decompose a Number into a Sum of Two Perfect Squares

5 Decomposing Numbers into Sums of Three Squares

Example 5.1

Is it possible to write 3929 as a sum of three squares? How about 2857? 3527?

There are 23 ways to decompose 3929 as a sum of three squares:

$$\begin{array}{lll}
 3929 = 0^2 + 35^2 + 52^2 & = 2^2 + 9^2 + 62^2 & = 2^2 + 26^2 + 57^2 \\
 = 2^2 + 30^2 + 55^2 & = 3^2 + 28^2 + 56^2 & = 5^2 + 40^2 + 48^2 \\
 = 6^2 + 7^2 + 62^2 & = 6^2 + 23^2 + 58^2 & = 7^2 + 42^2 + 46^2 \\
 = 8^2 + 12^2 + 61^2 & = 8^2 + 27^2 + 56^2 & = 9^2 + 22^2 + 58^2 \\
 = 12^2 + 43^2 + 44^2 & = 14^2 + 22^2 + 57^2 & = 16^2 + 37^2 + 48^2 \\
 = 20^2 + 35^2 + 48^2 & = 21^2 + 28^2 + 52^2 & = 22^2 + 23^2 + 54^2 \\
 = 22^2 + 41^2 + 42^2 & = 23^2 + 30^2 + 50^2 & = 27^2 + 40^2 + 40^2 \\
 = 28^2 + 29^2 + 48^2 & = 28^2 + 36^2 + 43^2 &
 \end{array}$$

There are 6 ways to decompose 2857 as a sum of three squares:

$$\begin{array}{lll}
 2857 = 0^2 + 16^2 + 51^2 & = 2^2 + 33^2 + 42^2 & = 3^2 + 12^2 + 52^2 \\
 = 16^2 + 24^2 + 45^2 & = 18^2 + 18^2 + 47^2 & = 18^2 + 33^2 + 38^2
 \end{array}$$

3527 cannot be written as a sum of three squares.

Proposition 5.2

$\forall n \in \mathbb{N}$, if $n \equiv 7 \pmod{8}$, then $\nexists a, b, c \in \mathbb{N}$ such that $n = a^2 + b^2 + c^2$.

Proof. Let $n \in \mathbb{N}$ satisfy $n \equiv 7 \pmod{8}$. Suppose for the sake of contradiction that $\exists a, b, c \in \mathbb{N}$ such that $n = a^2 + b^2 + c^2$. Taking the equation $\pmod{8}$, we obtain $7 \equiv a^2 + b^2 + c^2 \pmod{8}$. However, the only perfect squares in $\mathbb{Z}_8$ are 0, 1, and 4:

$$\begin{array}{l}
 0^2 \equiv 0 \pmod{8} \\
 1^2 \equiv 1 \pmod{8} \\
 2^2 \equiv 4 \pmod{8} \\
 3^2 \equiv 1 \pmod{8} \\
 4^2 \equiv 0 \pmod{8} \\
 5^2 \equiv 1 \pmod{8} \\
 6^2 \equiv 4 \pmod{8} \\
 7^2 \equiv 1 \pmod{8}.
 \end{array}$$

Therefore, $\forall a, b, c \in \mathbb{N}$, $a^2 + b^2 + c^2 \in \{0, 1, 2, 3, 4, 5, 6\} \pmod{8}$, completing the proof. $\square$

Example 5.3

Can the numbers $11,225,153 = 3929 \cdot 2857$ and $13,857,583 = 3929 \cdot 3527$ be written as a sum of three squares?

- Yes; in fact, $11,225,153 = 3929 \cdot 2857$ can be written as a sum of three squares in 604 ways.
- $13,857,583 = 3929 \cdot 3527$ cannot be written as a sum of three squares.

Conjecture 5.4. We assert, yet have not yet proven, $\forall n \in \mathbb{N}, \exists x, y, z \in \mathbb{N}$ such that $n = x^2 + y^2 + z^2$ iff $\nexists a, b \in \mathbb{N}$ such that $n = 4^a(8b + 7)$.

Remark 5.5. While it was true that the products of sums of two squares could be written as the sum of two squares themselves, this does not hold true for sums of three squares. Consider $3 = 1^2 + 1^2 + 1^2$ and $5 = 2^2 + 1^2 + 0^2$. It can be verified that their product, 15, cannot be written as a sum of three squares.

It is also not necessarily true that the product of two non-sums of three squares cannot be written as a sum of three squares. In fact, assuming that the previous conjecture is true,

Proposition 5.6

$\forall a_1, a_2, \dots, a_n \in \mathbb{N}$ that cannot be written as the sum of three squares, $a_1 a_2 a_3 \cdots a_n$ is a sum of three squares if n is even.

Proof. Consider two numbers $4^{a_1} \cdot b_1$ and $4^{a_2} \cdot b_2$, where $b_1, b_2 \equiv 7 \pmod{8}$ and $a_1, a_2, b_1, b_2 \in \mathbb{Z}^+$. Multiplying them together, we obtain

$$4^{a_1+a_2} \cdot b_1 b_2,$$

where $b_1 b_2 \equiv 49 \equiv 1 \pmod{8}$. By our previous conjecture, this product must be a sum of three squares.

Multiplying $4^{a_1+a_2} \cdot b_1 b_2$ by $4^{a_3} \cdot b_3$, where $a_3, b_3 \in \mathbb{Z}$, to obtain

$$4^{a_1+a_2+a_3} \cdot b_1 b_2 b_3,$$

where $b_1 b_2 b_3 \equiv 343 \pmod{8} \equiv 7 \pmod{8}$. By [Conjecture 5.4](#), this product cannot be a sum of three squares. As we continue to multiply by non-sums, we can see that the residue modulo 8 will cycle: first $1 \pmod{8}$, then $7 \pmod{8}$ again, and so on...

□

5.1 Code

```
1 import java.util.*;
2 public class ThreeSquares {
3     public static void main(String args[]) {
4         Scanner in = new Scanner(System.in);
5         //gets input from user
6         int x = in.nextInt();
7         //creates ArrayList to store decompositions
8         ArrayList<String> squares = new ArrayList<String>();
9         //loops over integers i, j less than sqrt(x), with i <= j
10        for (int i = 0; i < Math.sqrt(x); i++) {
11            for (int j = i; j <= Math.sqrt(x); j++) {
12                //calculates the number for which  $x = i^2 + j^2 + y^2$ 
13                double y = Math.sqrt(x-i*i-j*j);
14                //checks if y is an integer
15                if (Math.floor(y) == y && j <= y) {
16                    //adds decomposition to the ArrayList
17                    squares.add(i + " " + j + " " + Math.round(y));
18                }
19            }
20        }
21        //outputs the number of decompositions and each decomposition
22        System.out.println(squares.size());
23        for (String s:squares) {
24            System.out.println(s);
25        }
26    }
27 }
```

Listing 4: Counting the Number of Ways to Decompose a Number into a Sum of Three Perfect Squares

6 Decomposing Numbers into Sums of Four Squares

Example 6.1

Is it possible to write 3929 as a sum of four squares? How about 2857? 3527?

- There are 97 ways to decompose 3929 as a sum of four squares, one of which is $3929 = 28^2 + 30^2 + 33^2 + 34^2$.
- There are 73 ways to decompose 2857 as a sum of four squares, one of which is $2857 = 20^2 + 25^2 + 26^2 + 34^2$.
- There are 80 ways to decompose 3527 as a sum of four squares, one of which is $3527 = 25^2 + 27^2 + 27^2 + 38^2$.

Example 6.2

Can the numbers $11,225,153 = 3929 \cdot 2857$ and $13,857,583 = 3929 \cdot 3527$ be written as a sum of four squares?

- Yes. $11,225,153 = 3929 \cdot 2857$ can be written as the sum of four squares in 89855520 ways.
- Yes. $13,857,583 = 3929 \cdot 3527$ can be written as the sum of four squares in 110920320 ways.

Theorem 6.3

$\forall n \in \mathbb{N}, \exists a, b, c, d \in \mathbb{N}$ such that $n = a^2 + b^2 + c^2 + d^2$.

Proof. From [Conjecture 5.4](#), every natural number not of the form $n = 4^a(8b + 7)$ for $a, b \in \mathbb{N}$ can be written as a sum of three squares. If $a = 0$, then $n = 8b + 7 \equiv 3 \pmod{4}$, and if $a > 0$ then $n \equiv 0 \pmod{4}$. So, $\forall m \in \mathbb{N}$ satisfying $m \equiv 1$ or $2 \pmod{4}$, or $m \equiv 0 \pmod{4}$ but not of the form $n = 4^a(8b + 7)$, m can be written as a sum of three squares and thus four squares by adding 0^2 . Moreover, all $m \equiv 3 \pmod{4}$ can be written as a sum of four squares by adding 1^2 to the three squares that sum to $m - 1 \equiv 2 \pmod{4}$.

Now, the only numbers that have yet to be proven to be the sum of four squares are those of the form $4^a(8b + 7)$, where $a > 0$.

From above, we know that all positive integers of the form $8b + 7 \equiv 3 \pmod{4}$ can be written as the sum of four squares. Then $\exists w, x, y, z \in \mathbb{N}$ such that $8b + 7 = w^2 + x^2 + y^2 + z^2$. Multiplying this number by 4^a , we obtain

$$\begin{aligned} 4^a(w^2 + x^2 + y^2 + z^2) &= 4^a w^2 + 4^a x^2 + 4^a y^2 + 4^a z^2 \\ &= (2^a w)^2 + (2^a x)^2 + (2^a y)^2 + (2^a z)^2. \end{aligned}$$

This is a sum of four squares, thus completing the proof. □

6.1 Code

```
1 import java.util.*;
2 public class MyClass {
3     public static void main(String args[]) {
4         Scanner in = new Scanner(System.in);
5         //gets input from user
6         int x = in.nextInt();
7         ArrayList<String> squares = new ArrayList<String>();
8         //loops over all i, j, k less than sqrt(x), with i <= j <= k
9         for (int i = 0; i < Math.sqrt(x); i++) {
10             for (int j = i; j <= Math.sqrt(x); j++) {
11                 for (int k = j; k <= Math.sqrt(x); k++) {
12                     //calculates the number for which  $x = i^2 + j^2 + k^2 + y^2$ 
13                     double y = Math.sqrt(x - i*i - j*j - k*k);
14                     //checks if y is an integer
15                     if (Math.floor(y) == y && k <= y) {
16                         //adds the decomposition to the ArrayList
17                         squares.add(i + " " + j + " " + k + " " + Math.round(y));
18                     }
19                 }
20             }
21         }
22         //outputs the number of decompositions and each decomposition
23         System.out.println(squares.size());
24         for (String s:squares) {
25             System.out.println(s);
26         }
27     }
28 }
```

Listing 5: Counting the Number of Ways to Decompose a Number into a Sum of Four Perfect Squares

7 r -gonal Numbers

7.1 General Findings

Theorem 7.1

$\forall r, n \in \mathbb{Z}^+, r \geq 3$, the n th r -gonal number is

$$R_n = \frac{n(n-1)(r-2)}{2} + n. \quad (3)$$

Proof. We can approach this problem visually. To progress from the $(n-1)$ th r -gonal number to the n th r -gonal number, $r-2$ sides are being added to form a larger polygon, each with n dots. This gives a total of $n(r-2) - (r-3)$ new dots, since there are $r-3$ dots on two sides, thus being counted twice.

Similarly, to progress from the n th r -gonal number to the $n+1$ th r -gonal number, we add $(n+1)(r-2) - (r-3)$ new dots. Moreover, we can see that the difference of the differences between two consecutive r -gonal numbers is always $r-2$. This yields the formula

$$\begin{aligned} R_n &= 1 + (1 + (r-2)) + (1 + 2(r-2)) + \cdots + (1 + (n-1)(r-2)) \\ &= 1 + \sum_{i=1}^{n-1} (1 + i(r-2)) \\ &= 1 + (n-1) + (r-2) \sum_{i=1}^{n-1} i \\ &= n + (r-2) \cdot \frac{(n-1)(n-1+1)}{2} \\ &= \frac{n(n-1)(r-2)}{2} + n. \end{aligned}$$

□

Corollary 7.2

$$\forall r, n \in \mathbb{Z}^+, r \geq 3, R_n = \sum_{k=0}^{n-1} (r-2)k + 1.$$

Proof. Simplifying the expression quickly yields the result:

$$\begin{aligned}\sum_{k=0}^{n-1} (r-2)k + 1 &= (r-2) \sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 1 \\ &= (r-2) \frac{(n-1)(n)}{2} + n \\ &= R_n.\end{aligned}$$

□

Remark 7.3. Although the equation in [Corollary 7.2](#) is equivalent to that in [Theorem 7.1](#), this rewriting makes it easier to see that the n th r -gonal number is the sum of the first n positive integers satisfying $k \equiv 1 \pmod{r} - 2$.

Proposition 7.4

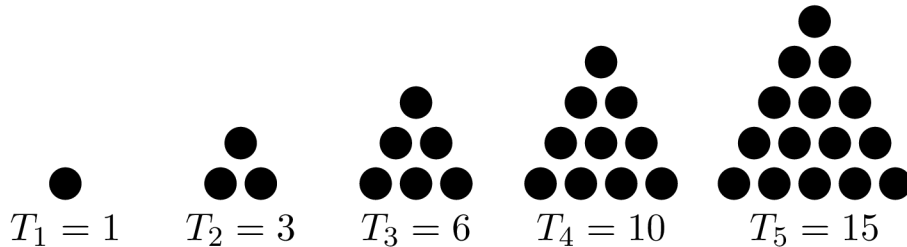
$\forall m, k \in \mathbb{Z}^+$ such that $\ell = \frac{m+k}{2} \in \mathbb{Z}$, $L_n = \frac{1}{2}(M_n + K_n)$.

Proof. Using [Theorem 7.1](#), we get

$$\begin{aligned}\frac{1}{2}(M_n + K_n) &= \frac{1}{2} \left(\frac{n(n-1)(m-2)}{2} + n + \frac{n(n-1)(k-2)}{2} + n \right) \\ &= \frac{1}{2} \left(\frac{n(n-1)(m+k-4)}{2} + 2n \right) \\ &= \frac{1}{2} \left(\frac{n(n-1)(2\ell-4)}{2} + 2n \right) \\ &= \frac{n(n-1)(\ell-2)}{2} + n \\ &= L_n.\end{aligned}$$

□

7.2 Triangular Numbers



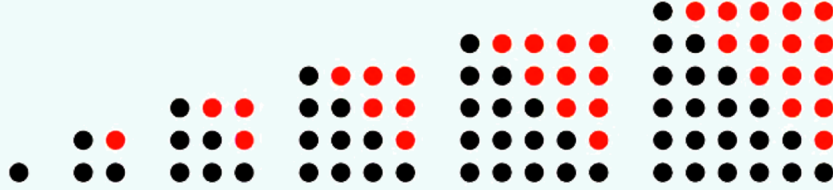
Corollary 7.5 $\forall n \in \mathbb{Z}^+,$

$$T_n = \frac{n(n+1)}{2}. \quad (4)$$

Proof. Using [Theorem 7.1](#) with $r = 3$, we get:

$$\begin{aligned} T_n &= \frac{n(n-1)(3-2)}{2} + n \\ &= \frac{n^2 - n + 2n}{2} \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

□

Proposition 7.6 $\forall n \in \mathbb{Z}^+, T_n + T_{n+1} = (n+1)^2.$ 

Proof. By [Corollary 7.4](#), $\forall n \in \mathbb{Z}^+$ we have:

$$\begin{aligned} T_n + T_{n+1} &= \frac{n(n+1)}{2} + \frac{(n+1)(n+2)}{2} \\ &= \frac{n^2 + n + n^2 + 3n + 2}{2} \\ &= n^2 + 2n + 1 \\ &= (n+1)^2. \end{aligned}$$

□

Theorem 7.7

$\forall n \in \mathbb{Z}^+, \exists x, y \in \mathbb{Z}^+$ such that $n = T_x + T_y$ iff $\exists a, b \in \mathbb{Z}$ such that $4n+1 = a^2 + b^2$.

Proof. Suppose $\exists x, y \in \mathbb{Z}^+$ such that $n = T_x + T_y = \frac{x(x+1)}{2} + \frac{y(y+1)}{2}$. Then we have:

$$\begin{aligned}
4n + 1 &= 4 \left(\frac{x(x+1)}{2} + \frac{y(y+1)}{2} \right) + 1 \\
&= 2x(x+1) + 2y(y+1) + 1 \\
&= 2x^2 + 2x + 2y^2 + 2y + 1 \\
&= (x^2 - 2xy + y^2) + (x^2 + 2x + y^2 + 2y + 2xy + 1) \\
&= (x - y)^2 + (x + y + 1)^2.
\end{aligned}$$

Thus, $4n + 1$ can be written as the sum of two squares.

Conversely, suppose that $4n + 1 = a^2 + b^2$. Since $4n + 1$ is odd, one of a, b must be even, and the other must be odd. Thus, both $a + b - 1$ and $a - b - 1$ are even. Let $c = \frac{a + b - 1}{2}$ and $d = \frac{a - b - 1}{2}$.

This yields

$$\begin{aligned}
\frac{c(c+1)}{2} + \frac{d(d+1)}{2} &= \frac{1}{2} \left(\frac{a+b-1}{2} \cdot \frac{a+b+1}{2} + \frac{a-b-1}{2} \cdot \frac{a-b+1}{2} \right) \\
&= \frac{1}{8} ((a+b-1)(a+b+1) + (a-b-1)(a-b+1)) \\
&= \frac{1}{8} ((a+b)^2 - 1 + (a-b)^2 - 1) \\
&= \frac{1}{8} (2a^2 + 2b^2 - 2) \\
&= \frac{1}{4} (a^2 + b^2 - 1) \\
&= \frac{1}{4} (4n + 1 - 1) \\
&= n.
\end{aligned}$$

Therefore, if $4n + 1$ can be written as the sum of two squares, n is the sum of two triangular numbers. $\square$

Theorem 7.8

$\forall n \in \mathbb{Z}^+, \exists x, y, z \in \mathbb{N}$ such that $n = T_x + T_y + T_z$.

Proof. From [Conjecture 5.4](#), an integer $8n + 3 \equiv 3 \pmod{8}$ can be written as a sum of three squares, since it is not of the form $4^a(8b + 7)$. Thus, $\exists a, b, c \in \mathbb{N}$ such that $8n + 3 = a^2 + b^2 + c^2$. As shown in [Proposition 5.2](#), the only quadratic residues modulo

8 are 0, 1, and 4, so $a^2, b^2, c^2 \equiv 1 \pmod{8}$. This means that a, b, c are all odd, so $\exists x, y, z \in \mathbb{N}$ such that $a = 2x + 1$, $b = 2y + 1$, and $c = 2z + 1$. This yields:

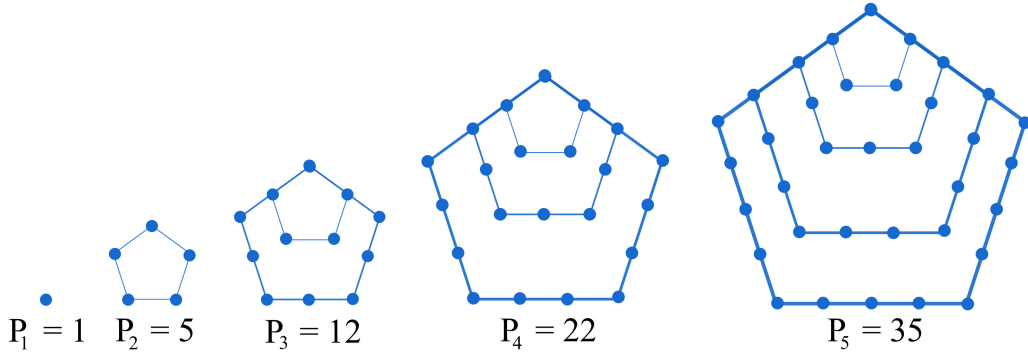
$$\begin{aligned}
 8n + 3 &= (2x + 1)^2 + (2y + 1)^2 + (2z + 1)^2 \\
 &= 4x^2 + 4x + 1 + 4y^2 + 4y + 1 + 4z^2 + 4z + 1 \\
 &= 4(x^2 + x + y^2 + y + z^2 + z) + 3 \\
 &= 4(x(x + 1) + y(y + 1) + z(z + 1)) + 3 \\
 n &= \frac{x(x + 1)}{2} + \frac{y(y + 1)}{2} + \frac{z(z + 1)}{2} \\
 &= T_x + T_y + T_z.
 \end{aligned}$$

□

Remark 7.9. The process used above demonstrates that finding a decomposition of n into three triangular numbers is equivalent to finding a decomposition of $8n + 3$ into three perfect squares.

Remark 7.10. Using [Theorem 7.1](#) with $r = 4$ gives $\frac{n(n - 1)(4 - 2)}{2} + n = n(n - 1) + n = n^2$, indeed yielding the perfect squares.

7.3 Pentagonal Numbers



Proposition 7.11

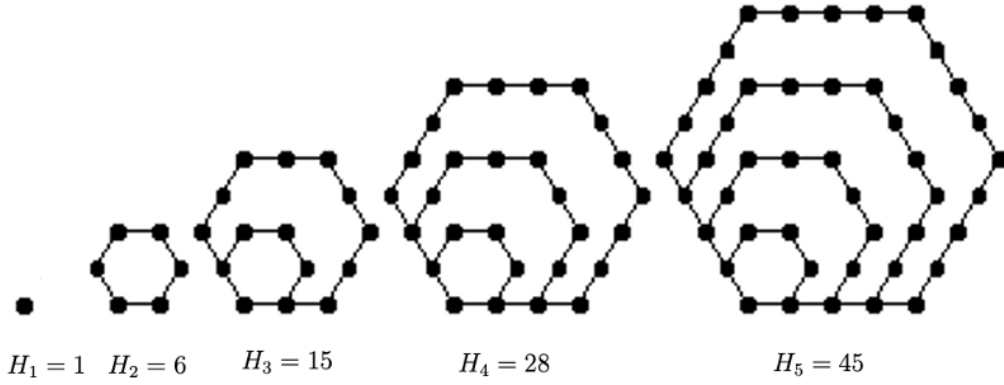
$\forall n \in \mathbb{Z}^+, 3P_n = T_{3n-1}$.

Proof. Using [Theorem 7.1](#) with $r = 5$, we can calculate $3P_n$:

$$\begin{aligned}
 3P_n &= 3 \left(\frac{n(n-1)(5-2)}{2} + n \right) \\
 &= \frac{9n^2 - 9n}{2} + 3n \\
 &= \frac{9n^2 - 3n}{2} \\
 &= \frac{3n(3n-1)}{2} \\
 &= T_{3n-1}.
 \end{aligned}$$

□

7.4 Hexagonal Numbers



Proposition 7.12

$\forall n \in \mathbb{Z}^+, H_n = T_{2n-1}.$

Proof. Using [Theorem 7.1](#) with $r = 6$, we can calculate H_n :

$$\begin{aligned}
 H_n &= \frac{n(n-1)(6-2)}{2} + n \\
 &= \frac{4n^2 - 4n}{2} + n \\
 &= \frac{4n^2 - 2n}{2} \\
 &= \frac{2n(2n-1)}{2} \\
 &= T_{2n-1}.
 \end{aligned}$$

□

7.5 Numerical Examples

Decompositions into at most r r -gonal numbers				
r	Triangular	Square	Pentagonal	Hexagonal
8	1+1+6	4 + 4	1+1+1+5	1+1+6
37	1+36	1+36	1+1+35	1+1+1+6+28
44	6+10+28	4+4+36	5+5+12+22	1+15+28
70	1+3+66	1+1+4+64	70	1+1+1+1+66
91	91	1+9+81	5+5+21+70	91

Conjecture 7.13. We assert, yet have not yet proven, that given $n \in \mathbb{Z}^+$, $\forall r \in \mathbb{Z}, r \geq 3$, n can be written as the sum of at most r r -gonal numbers.

8 Conclusion

In this paper, we showed that sums of squares have diverse applications in mathematics. Considering a lattice grid, the number of distinct distances at first appears to only bear some connections to the sequence of triangular numbers, but as issues with overcounting arise, we soon come to realize that sums of squares become relevant in tackling the problem of how many distinct distances lie in an $n \times n$ grid. We also explored another case, which similarly relates a geometric problem to of sums of squares, focuses on counting the number of lattice points within a closed disk. Here, we counted the number of ways in which distances, taken from the center of the circle, with a maximum length r , can be decomposed into sums of two integral squares. We also discovered that we can decompose every element in the set of natural natural numbers into some integral sum of squares by studying the patterns of their prime divisors. In this paper, we also showed the utility of various properties in the Gaussian integers in answering questions about sums of squares. Finally, we explored the connections between sums of squares and r -gonal numbers, delving into an exploration of the triangular numbers through further experimentation with pentagonal and hexagonal numbers, we were able to make a remarkable conjecture about the sums of general r -gonal numbers. Although we were able to answer many of our probing questions in this paper, many more questions remain, requiring more advance methods than those used here.

References

- [1] “The First Six Triangular Numbers (Not Starting with T0).” Wikipedia, https://en.wikipedia.org/wiki/Triangular_number.
- [2] “Figure Showing the First Five Pentagonal Numbers.” Web.Net, <https://www.math.net/pentagonal-number>.
- [3] “Figure Showing the First Five Hexagonal Numbers.” Topsider, <http://topcoder.bgcoder.com/print.php?id=627>.